Hwang 2022 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — batch mixing & continuity → GO

38 retained patients intermix on one continuous manifold (same-sample neighbour fraction 0.96→0.20 across the θ sweep, near the mixing floor); radius-graph leaves one connected component.

Gate 2 — embryonic module geography → NO-GO

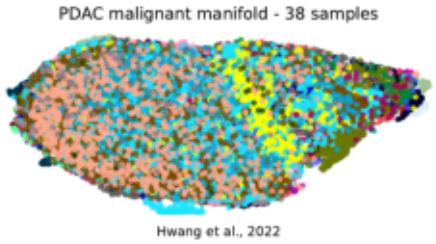
Pattern ABSENT. Three mutually-exclusive territories, not a hub→arms axis: all modules pairwise anti-correlated (Gast/pSom −0.33, pSom/Crest −0.39, Gast/Crest −0.30) and no two co-occur (all joint-high enrichments depleted → no convergence hub). Crest mid-axis (decile pt 0.50), NOT terminal.

Gate 3 — hub cross-repression → NA

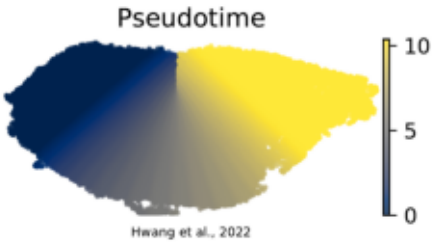
Not branching and Gate 2 NO-GO — cross-repression test not applicable.

STEP FIGURES

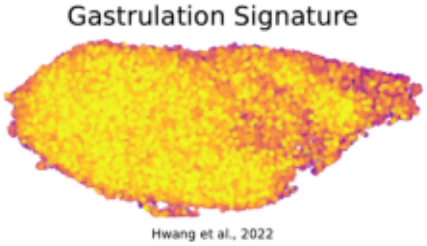
Gate 1: integration



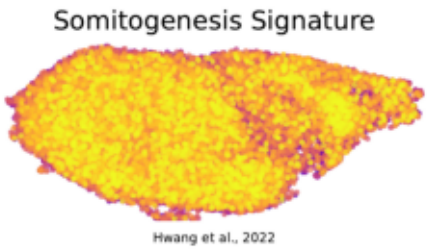
Trajectory / root



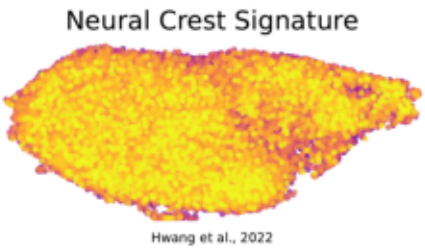
Gast



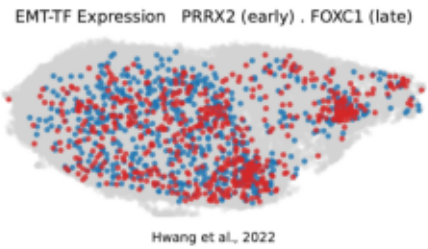
pSom



Crest



EMT-TFs



Dataset. Pancreatic ductal adenocarcinoma (Hwang et al., 2022), 10x; 64,538 malignant cells across 38/43 samples (≥14-cell threshold). Whole malignant manifold; cell cycle flat across lobes.

Verdict. ✗ Embryonic pattern does NOT reproduce (Gate 1 GO / Gate 2 NO-GO / Gate 3 NA). Modules resolve into three exclusive non-developmental territories with no terminal Crest readout — the Peng2019 same-PDAC NO-GO signature; territories track EMT-III / G2-M / classical states, not one developmental axis.

Parameter choices

Min cells/sample	14 (mult-gap 24→9, ratio 2.67; 38/43)	**Integration**	Harmony θ=10 on sample; n_pcs=20
Scope	whole manifold (cell cycle flat)	**Trajectory**	scFates curve N=20 (PPT rejected)
Root node	Gast-high tip (developmental origin)	**EMT-TFs**	PRRX2 (early) · FOXC1 (late)